



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Threats and Mitigation.

TASK 1.

❖ Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

➤ Common Network Threats:

Network Threat	Real-World Example
1. Phishing	A fake Instagram login link may trick a user into entering their username and password.
2. DDoS Attack	Attackers may send huge amounts of traffic to an online service, making it slow or unavailable.
3. Man-in-the-Middle (MITM)	An attacker on an unsafe public Wi-Fi network may try to intercept a user's data while using an app.
4. Malware	A user may download a fake WhatsApp or Instagram file that secretly installs harmful software.
5. Password Attack	Attackers may repeatedly guess a weak Paytm password to gain unauthorized account access.

TASK 2.

- ❖ Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value, The script should print the number of attempts it took to guess the correct PIN.

➤ Brute-Force Attack Simulation:

Python Code:

```
preset_pin = "5724"
attempts = 0

for i in range(10000):
    pin = f"{i:04d}"
    attempts += 1

    if pin == preset_pin:
        print("PIN guessed:", pin)
        print("Number of attempts:", attempts)
        break
```

Output:

```
PIN guessed: 5724
Number of attempts: 5725
```

Result: The script checks every possible 4-digit combination until it finds the correct PIN and shows the total number of attempts.

TASK 3.

❖ **List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

➤ **Physical Security Risks:**

Physical Security Risk	Mitigation Step
1. Unauthorized access	Use biometric locks and CCTV cameras to control and monitor entry.
2. Fire	Install smoke detectors and automatic fire-suppression systems.
3. Power failure	Use UPS and backup generators to keep servers running.
4. Overheating	Maintain proper cooling systems and temperature monitoring in the server room.

TASK 4.

- ❖ Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

➤ Physical Security Floor Plan:

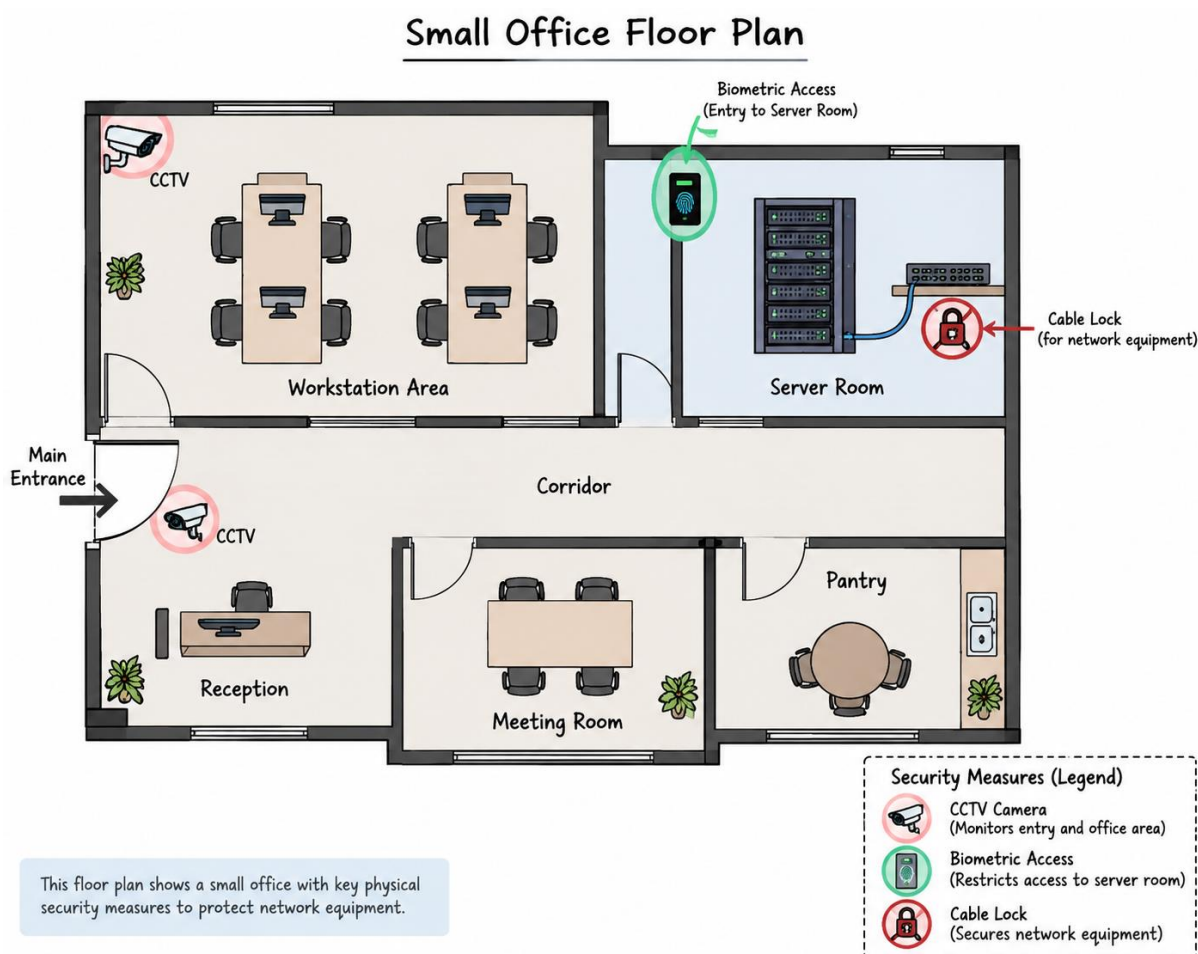


Figure : Small Office Floor Plan.

The floor plan shows three physical security measures for protecting network equipment:

- ⌘ **CCTV Camera** – Monitors the office and entry areas.

- ✂ **Biometric Access** – Restricts entry to the server room.
- ✂ **Cable Lock** – Secures the network equipment from physical theft.

TASK 5.

❖ **Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

➤ **Security Checklist for College Computer Lab:**

No.	Action
1.	Keep all lab computers and network devices updated with security patches.
2.	Use strong passwords and user accounts to prevent unauthorized access.
3.	Install and regularly update antivirus and firewall protection.
4.	Restrict access to the server room and network equipment using locks or biometric access.
5.	Install CCTV cameras to monitor the lab and equipment areas.
6.	Secure network devices and computers with cable locks to prevent physical theft.
7.	Regularly monitor network activity and inspect physical equipment for suspicious changes or damage.